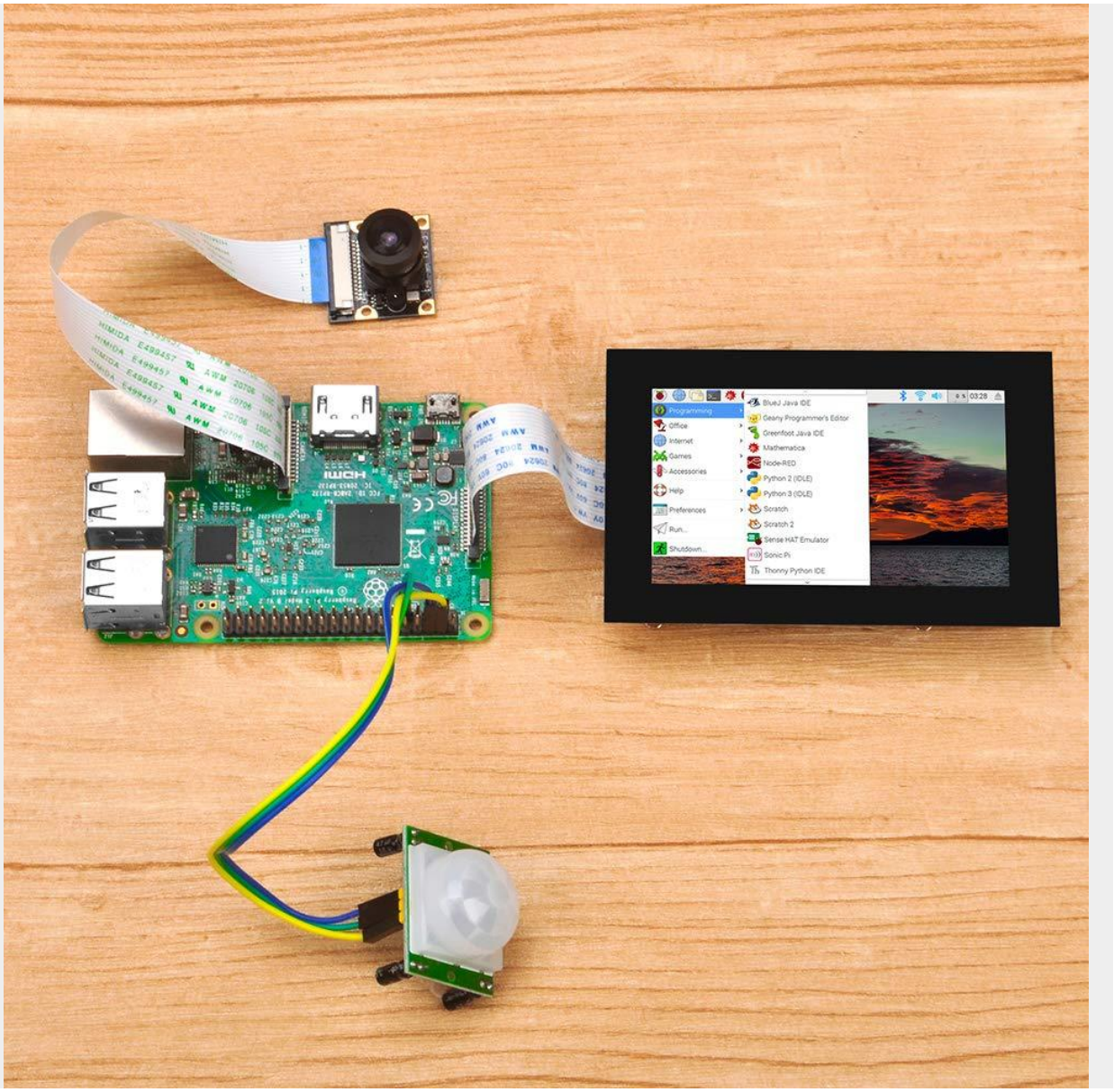
50 Raspberry Pi 5 projects using motion sensor, and display sensor

1. Security and surveillance projects



1.Motion-Activated Security Camera
Detect movement and display live camera footage on an LCD.



2. Smart Intruder Alarm
Display intruder alerts, motion status, and alarm state.



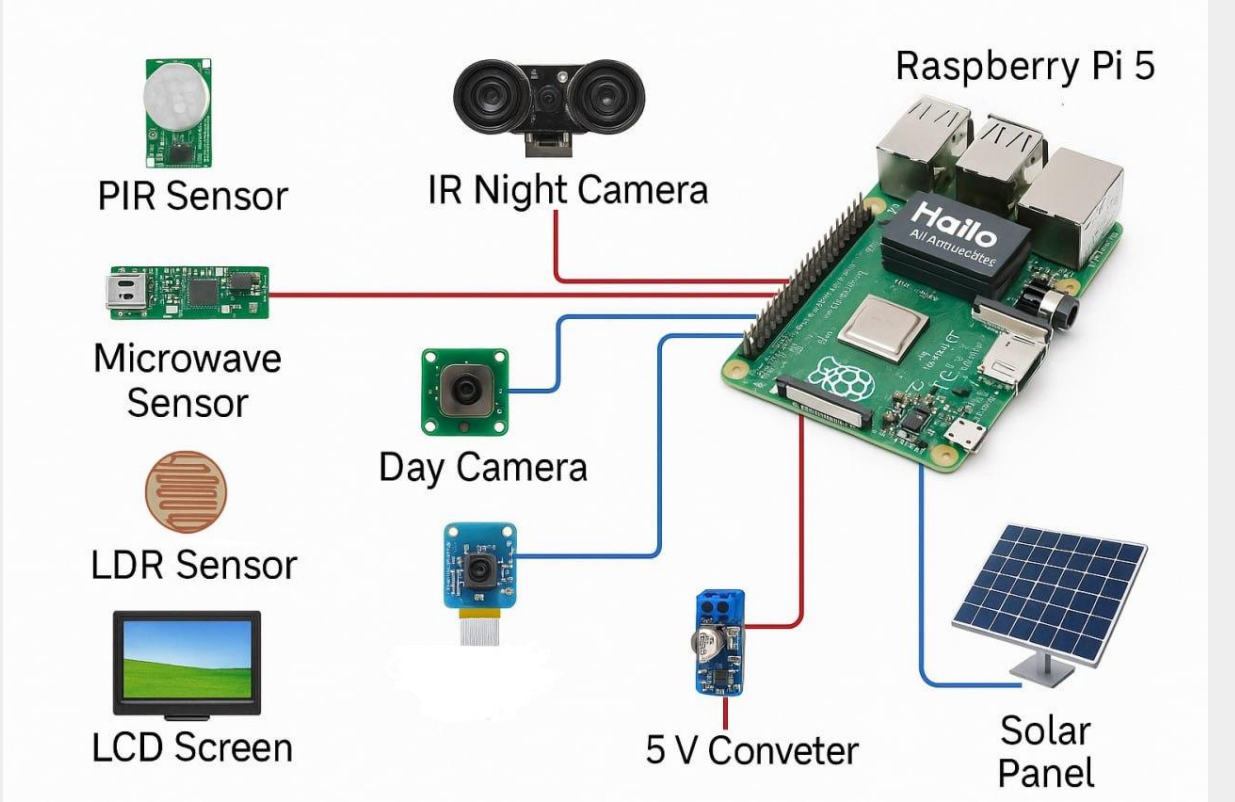
3. Smart Door Monitoring System
Show when someone approaches or opens a monitored entrance.



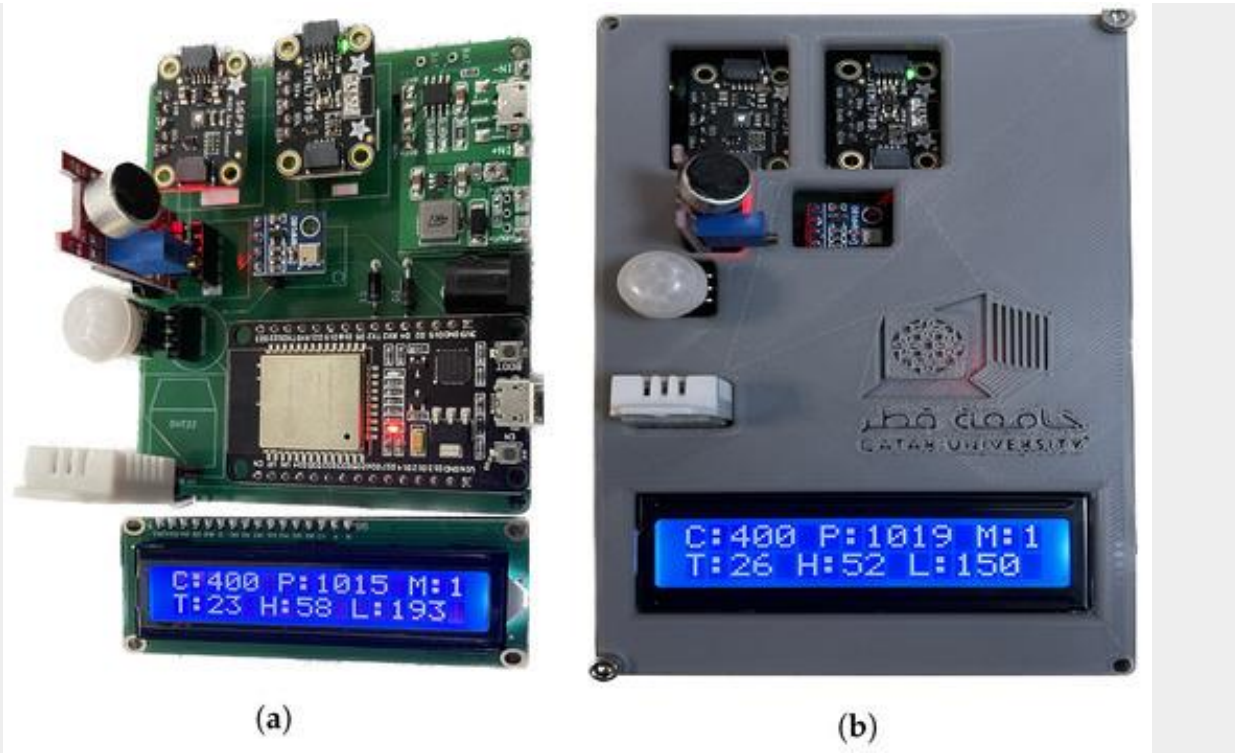
4. Home Security Dashboard
Display motion events, timestamps, and sensor status.



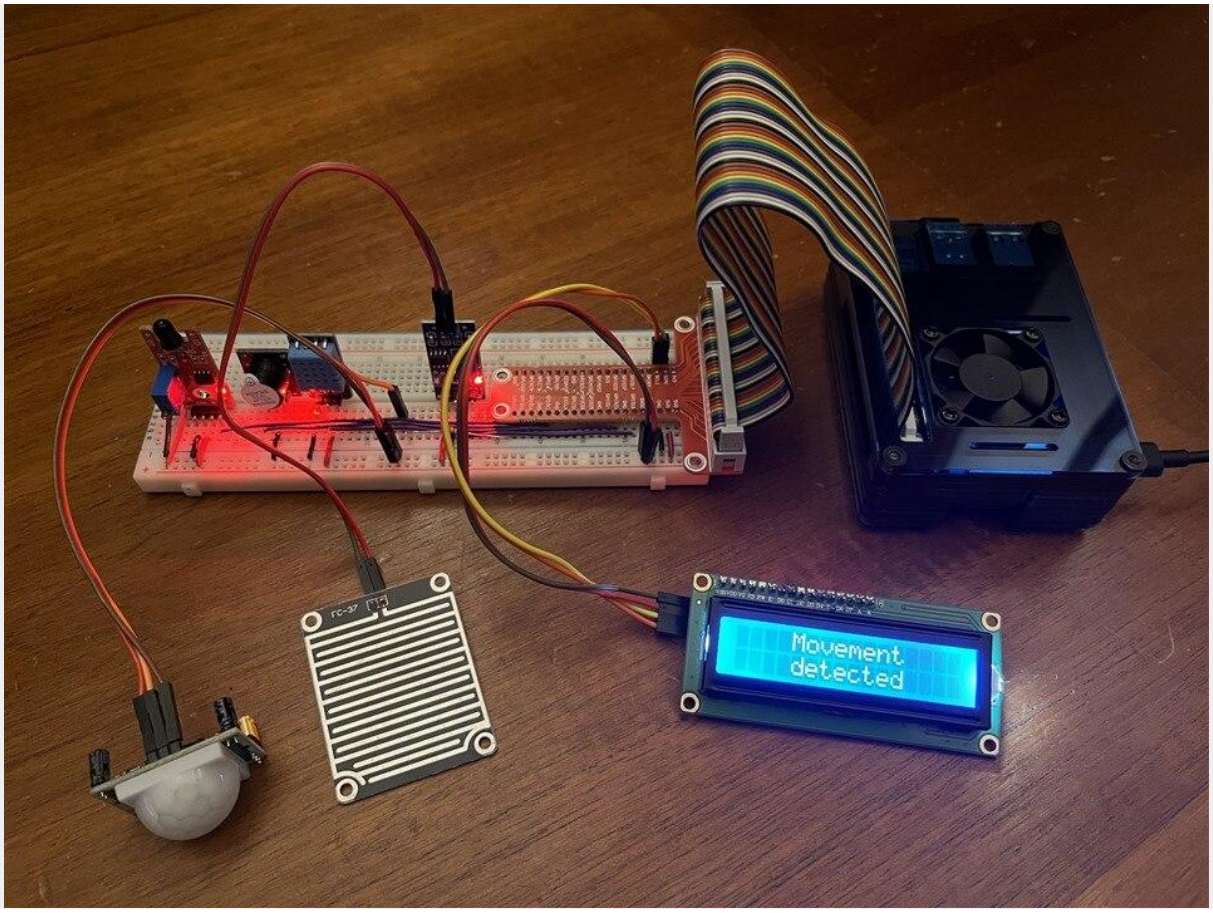
5. Smart Mailbox Detector
Detect mailbox activity and display delivery notifications.



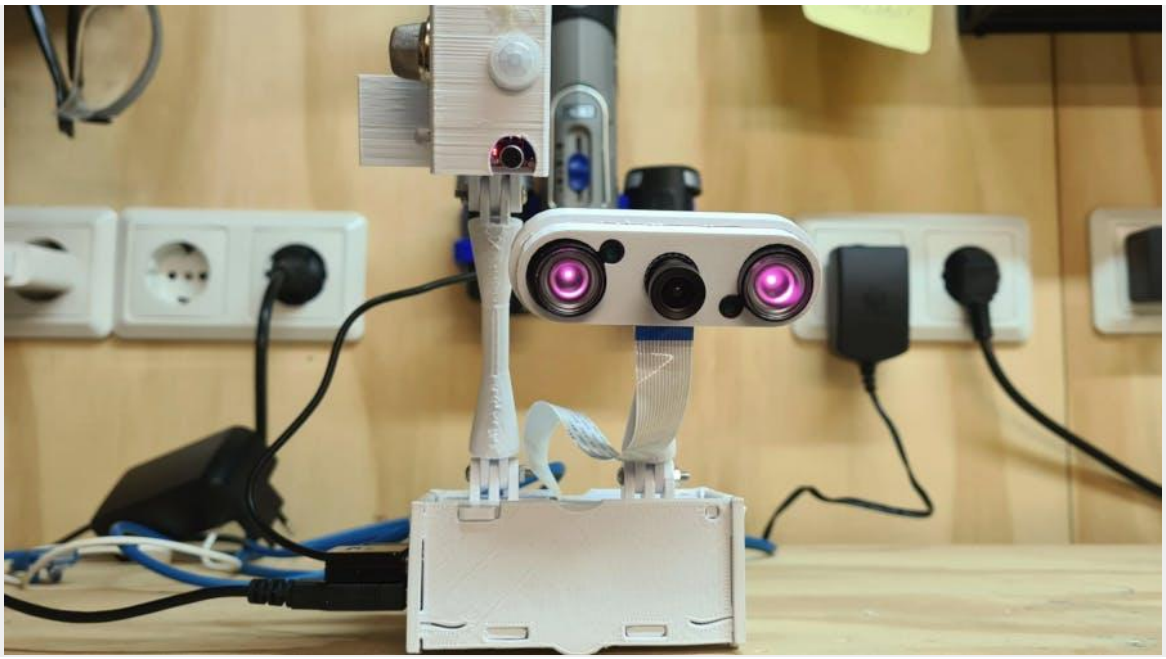
6. Wildlife Motion Camera
Detect animals and display captured images.



7. Room Occupancy Monitor
Display whether a room is occupied or vacant.



8. Garage Motion Monitor
Show garage activity and motion detection history.



9. Baby Room Activity Monitor
Display room activity and motion alerts.



10. Restricted Area Alert System
Display warnings when motion is detected in a restricted area.

2. Smart home automation projects

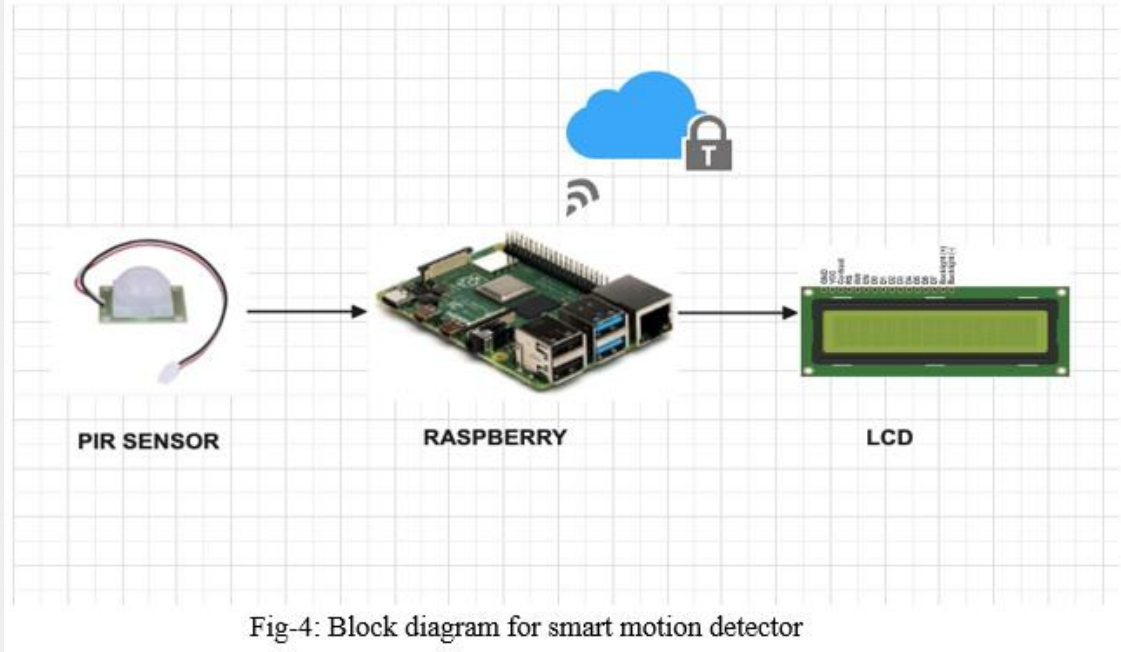
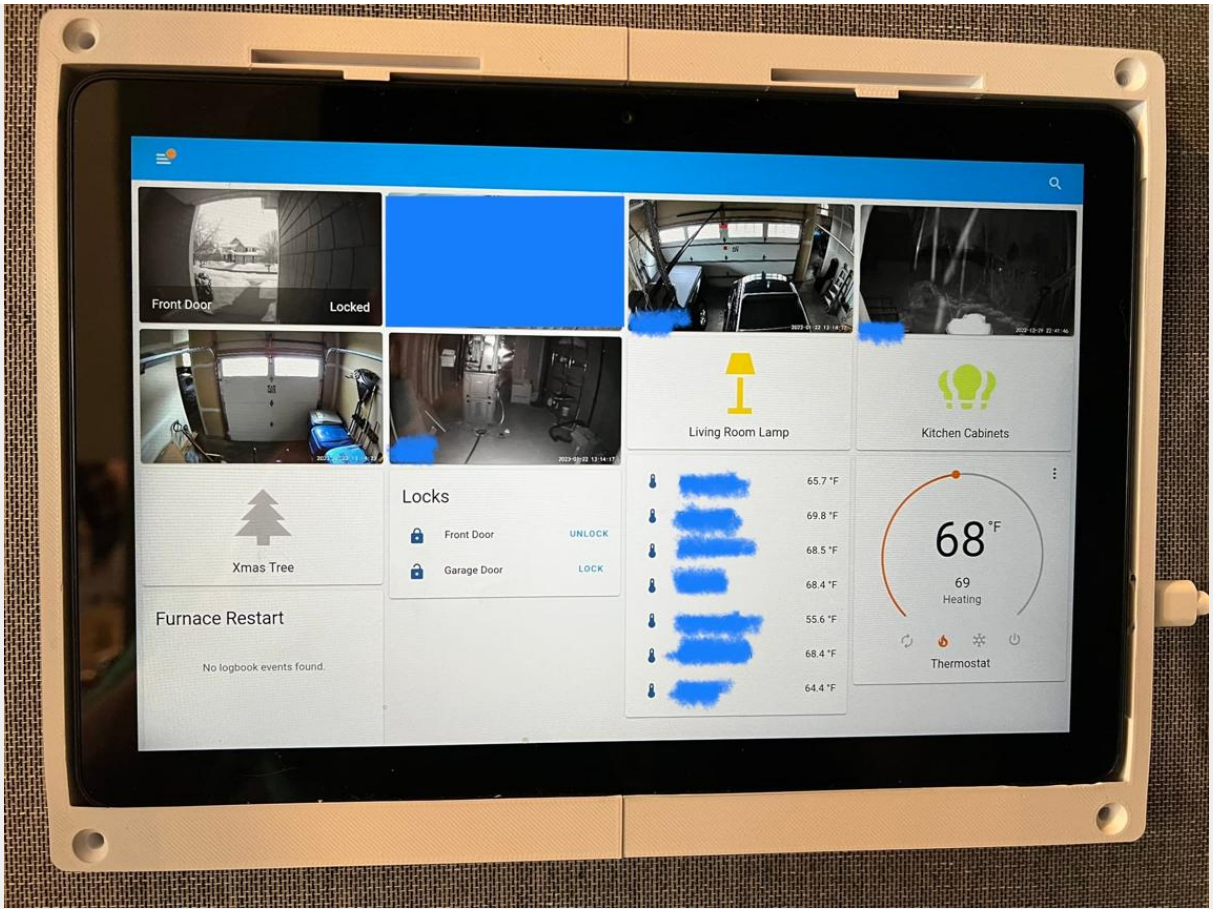


Fig-4: Block diagram for smart motion detector

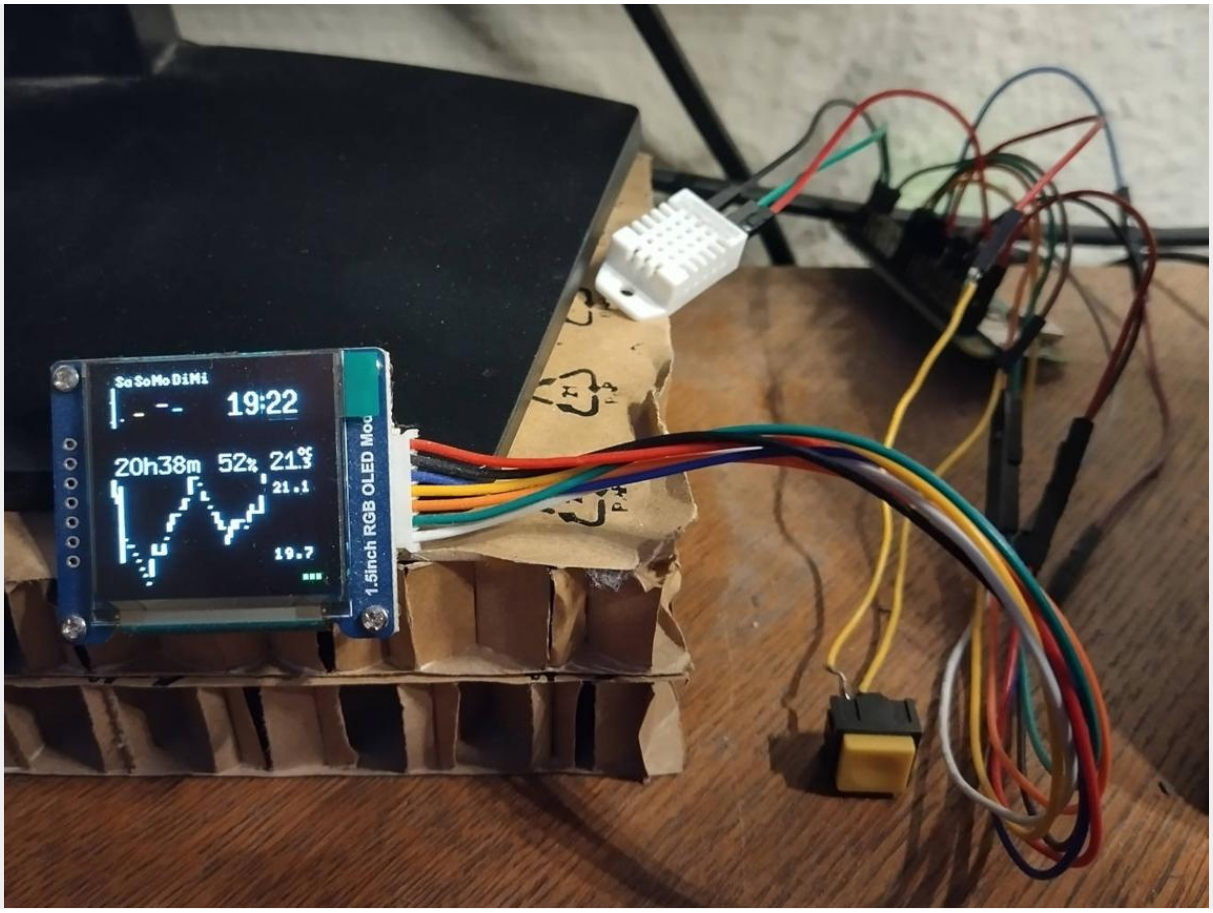
11. Automatic Room Lighting
Turn lights on when motion is detected and show light status.



12. Smart Home Control Panel
Display motion, lights, temperature, and appliance status.



13. Motion-Based Fan Controller
Activate a fan when someone enters a room.



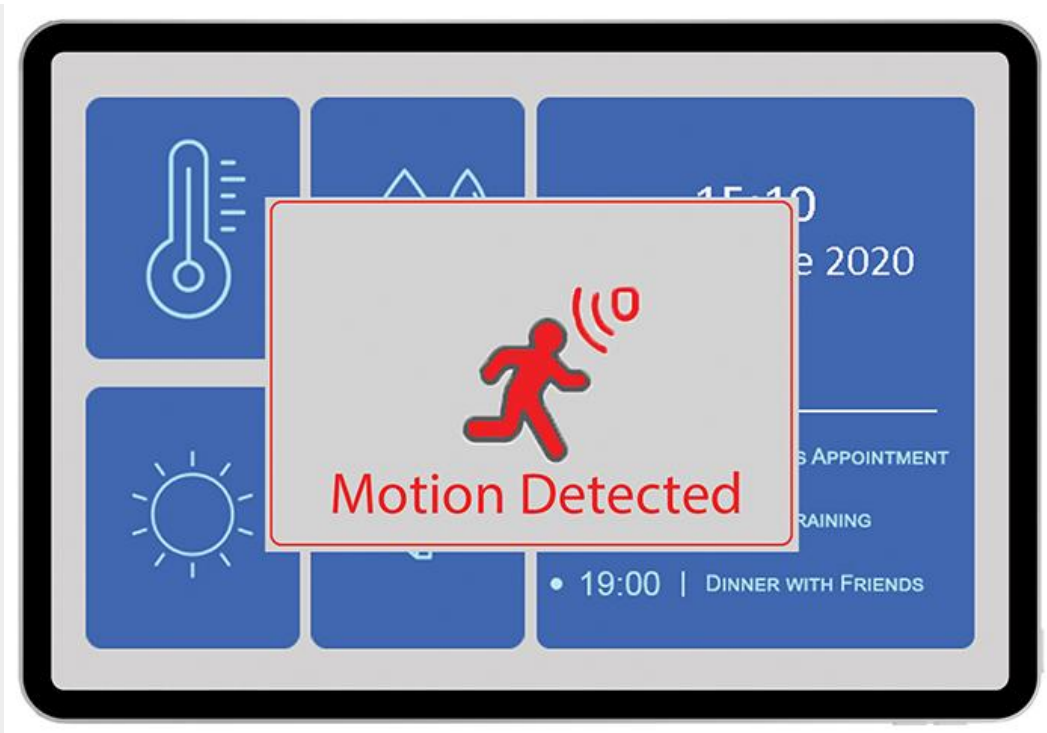
14. Smart Night Light
Turn on dim lighting when movement is detected at night.



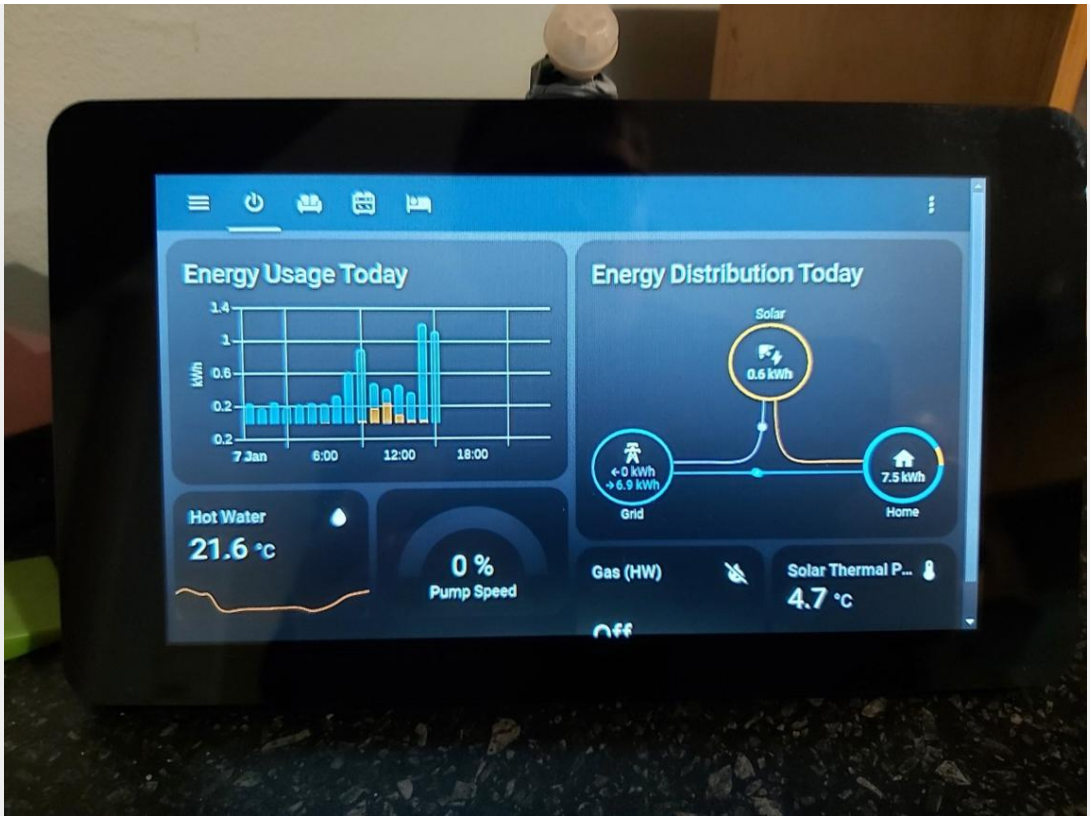
15. Automatic Curtain Controller
Display curtain status and trigger opening or closing.



16. Smart Bathroom Controller
Automatically control lighting and ventilation.



17. Smart Kitchen Assistant
Display cooking timers and activate the screen when someone approaches.



18. Energy-Saving Room Monitor
Display occupancy and estimated energy savings from automatic lighting.



19. Smart Staircase Lighting
Detect movement and activate staircase lights.



20. Smart Office Occupancy System
Display desk or room availability.

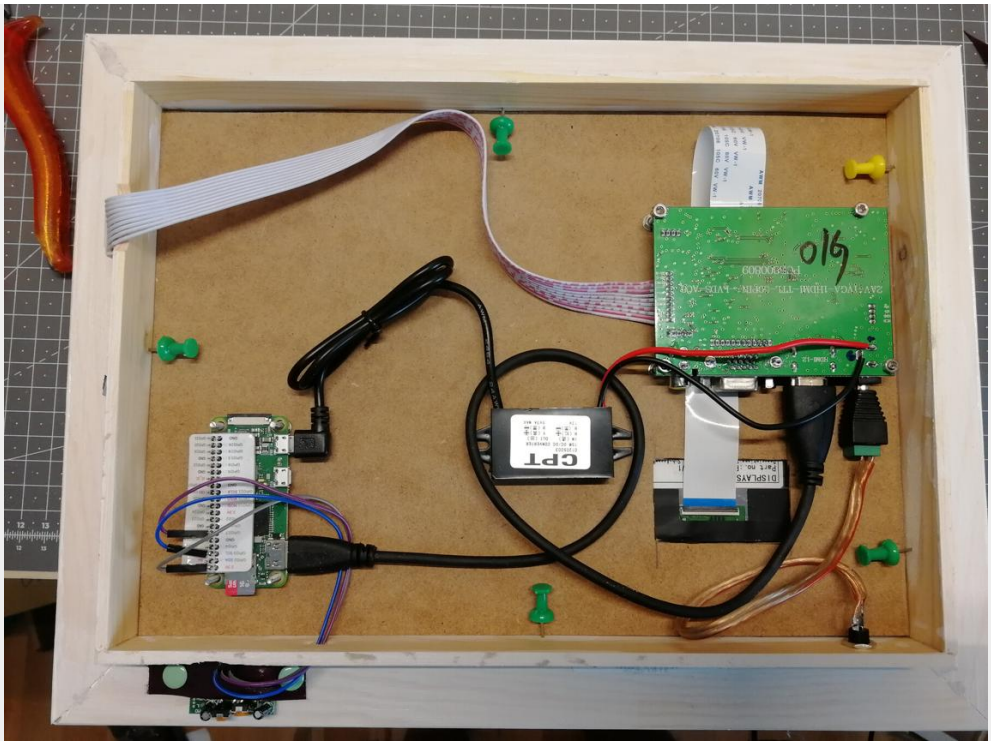
3. Interactive display and information projects



21. Motion-Activated Smart Mirror
Display time, weather, calendar, and news when someone approaches.



22. Interactive Information Kiosk
Wake the touchscreen when a visitor approaches.



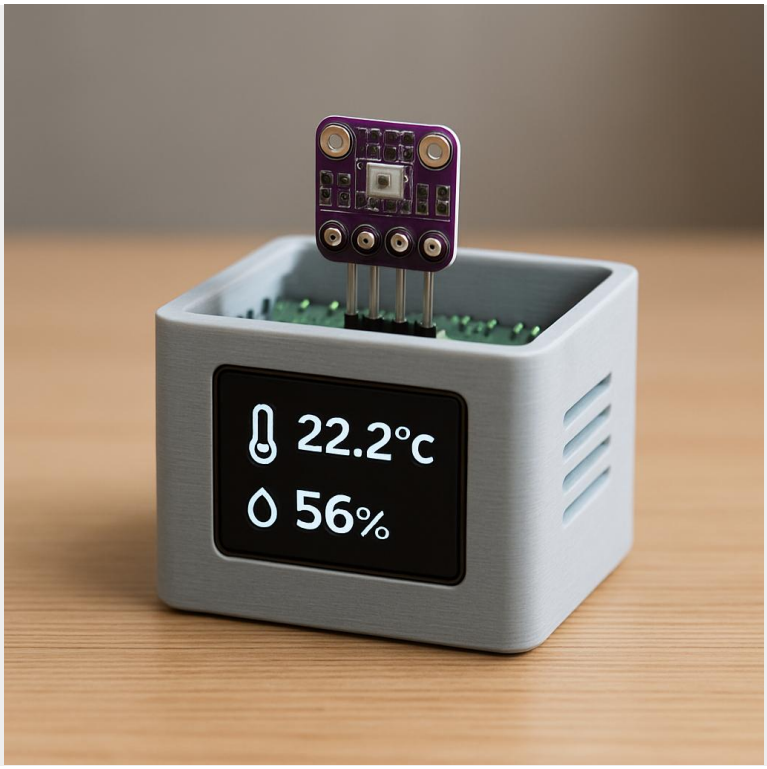
23. Smart Digital Photo Frame
Show pictures only when someone is nearby.



24. Smart Welcome Display
Show welcome messages when a person enters.



25. Motion-Activated Digital Calendar
Display schedules and reminders when approached.



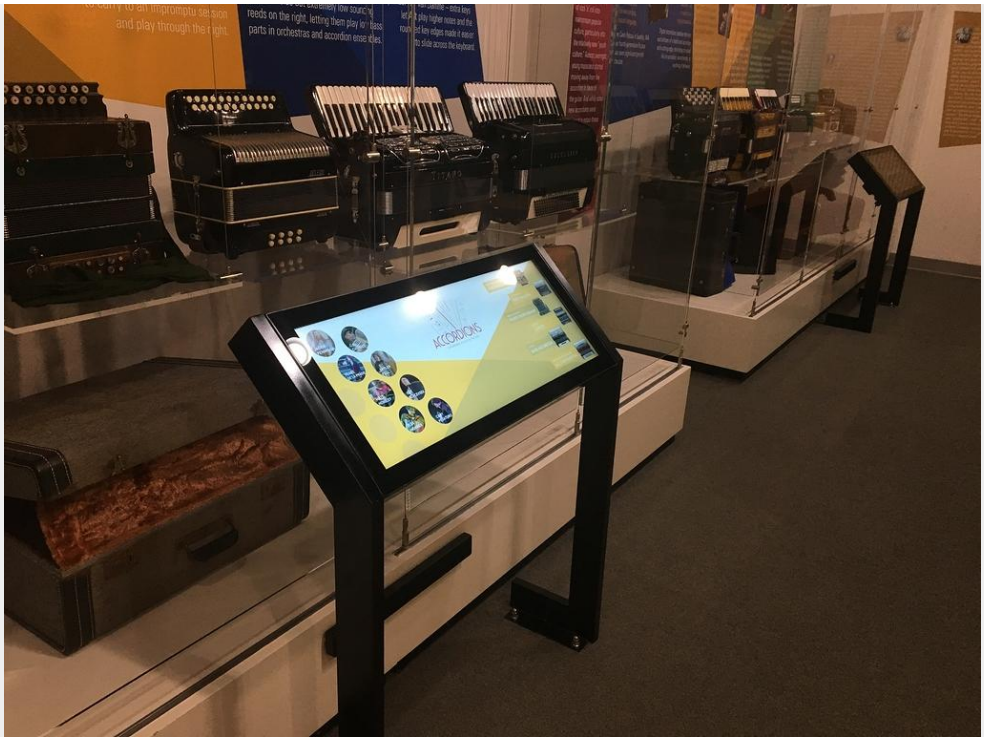
26. Smart Weather Station
Wake the display on motion and show weather information.



27. Motion-Activated Smart Clock
Display time, date, and alarms when someone approaches.



28. Smart Personal Assistant
Display reminders, tasks, and information on demand.



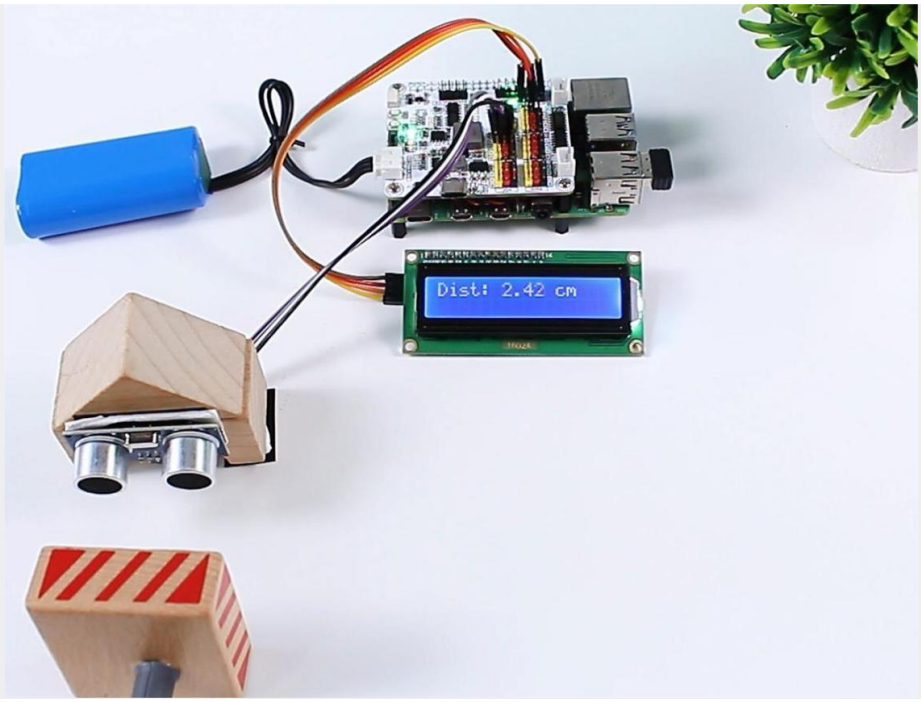
29. Interactive Museum Display
Show information about an exhibit when visitors approach.



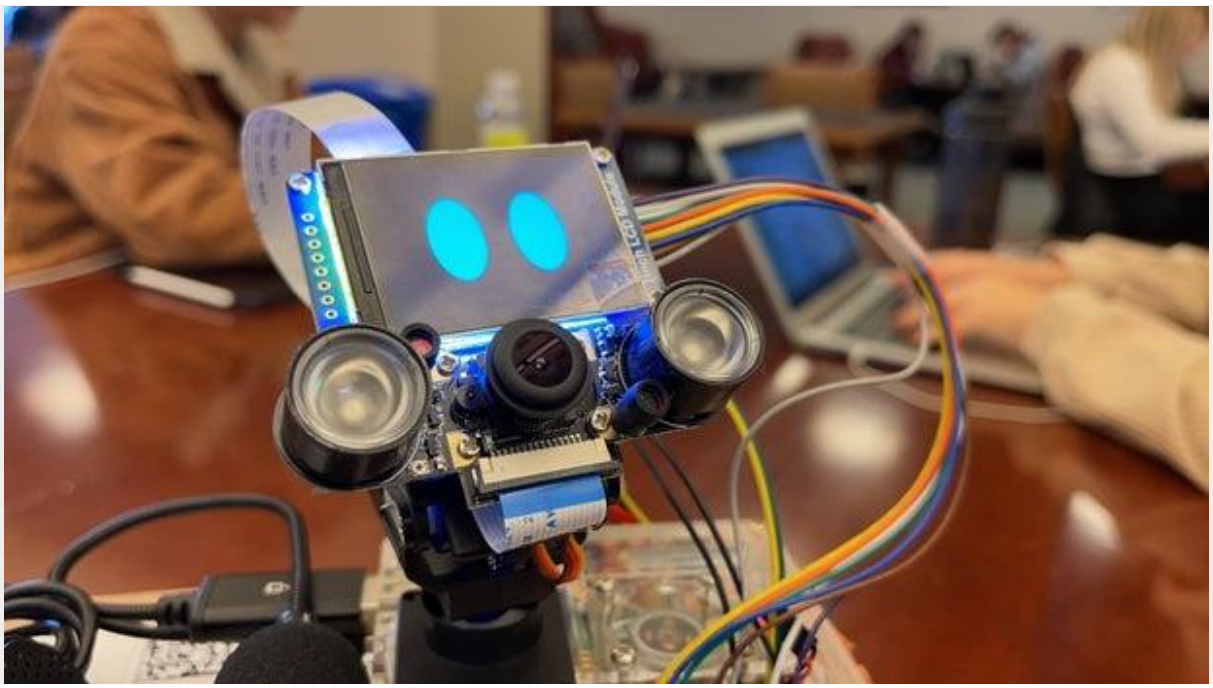
30. Motion-Activated Pixel Art Display
Display animations when motion is detected.

4. Robotics and engineering projects

Projects 31–40



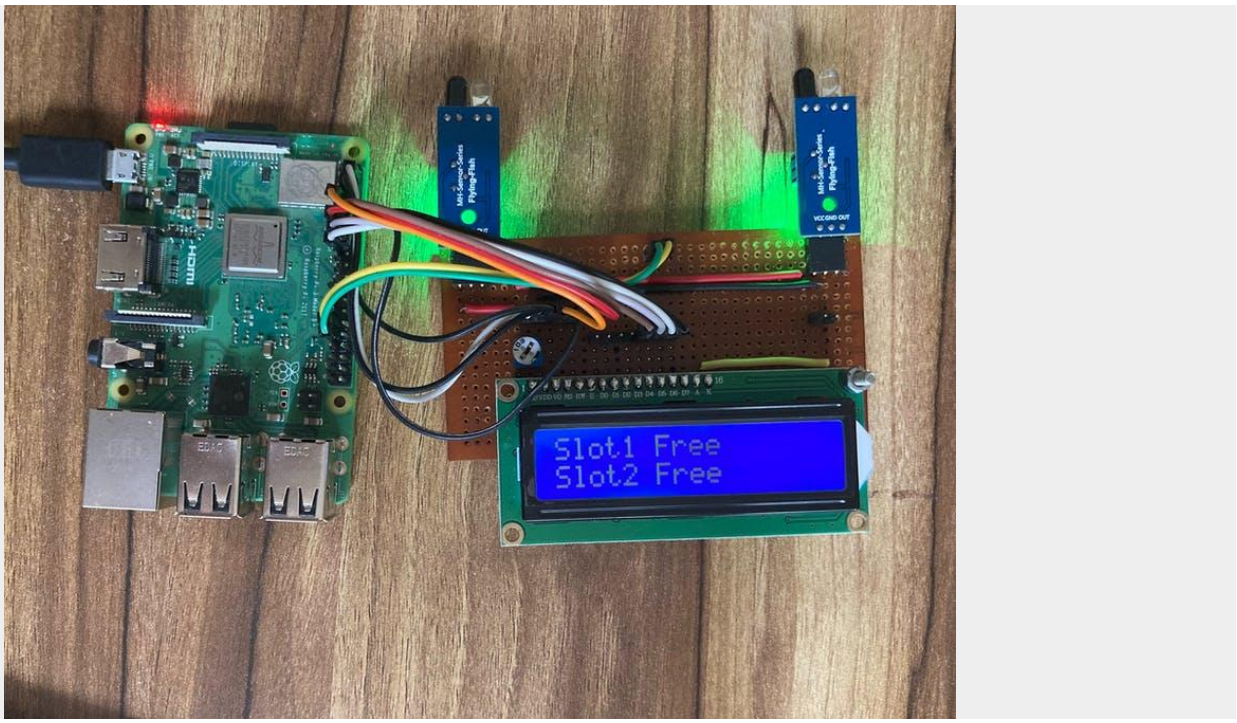
31. Motion-Detecting Robot
Display detected movement and robot status.



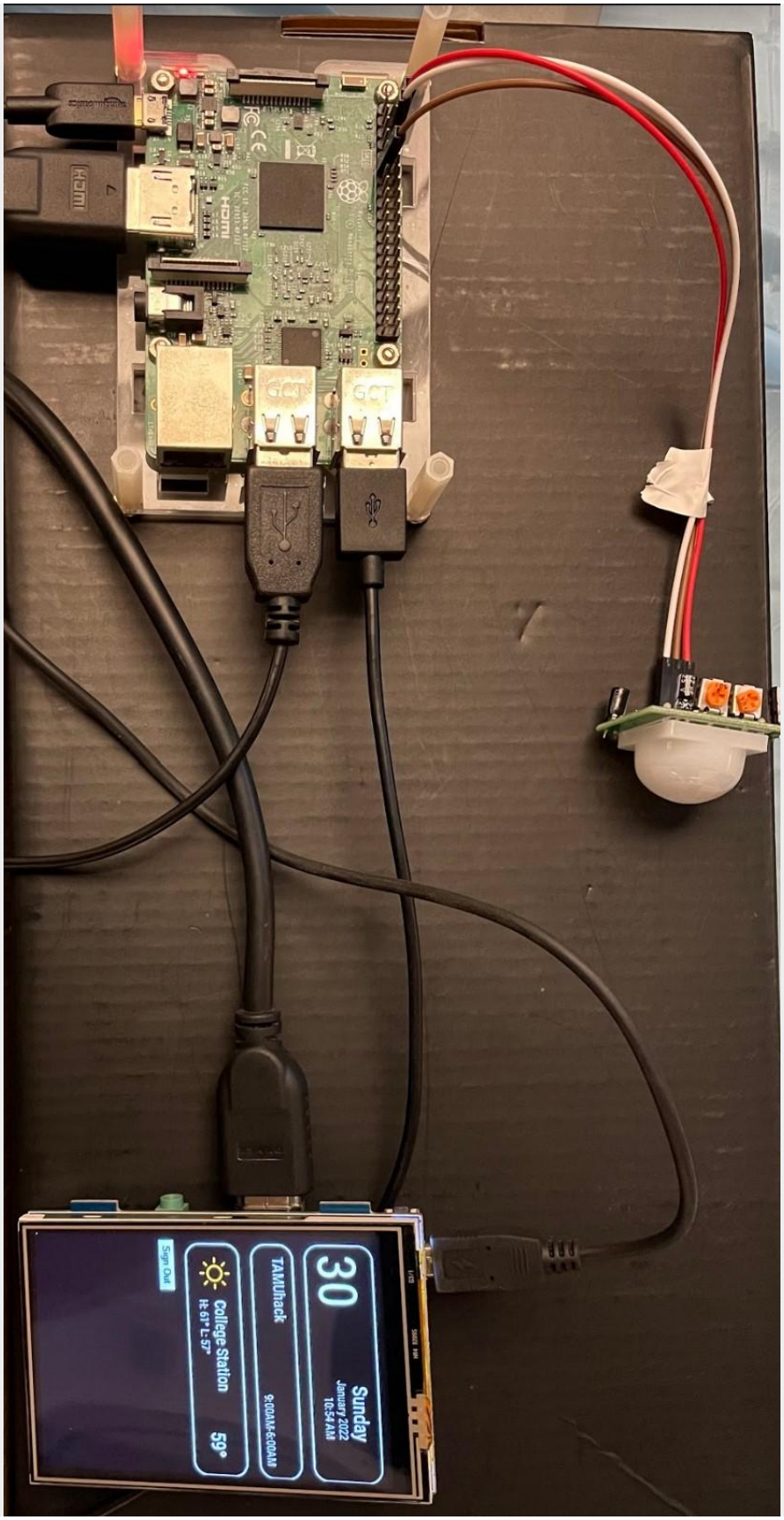
32. Motion-Tracking Robotic Head
Detect nearby movement and turn toward the person.



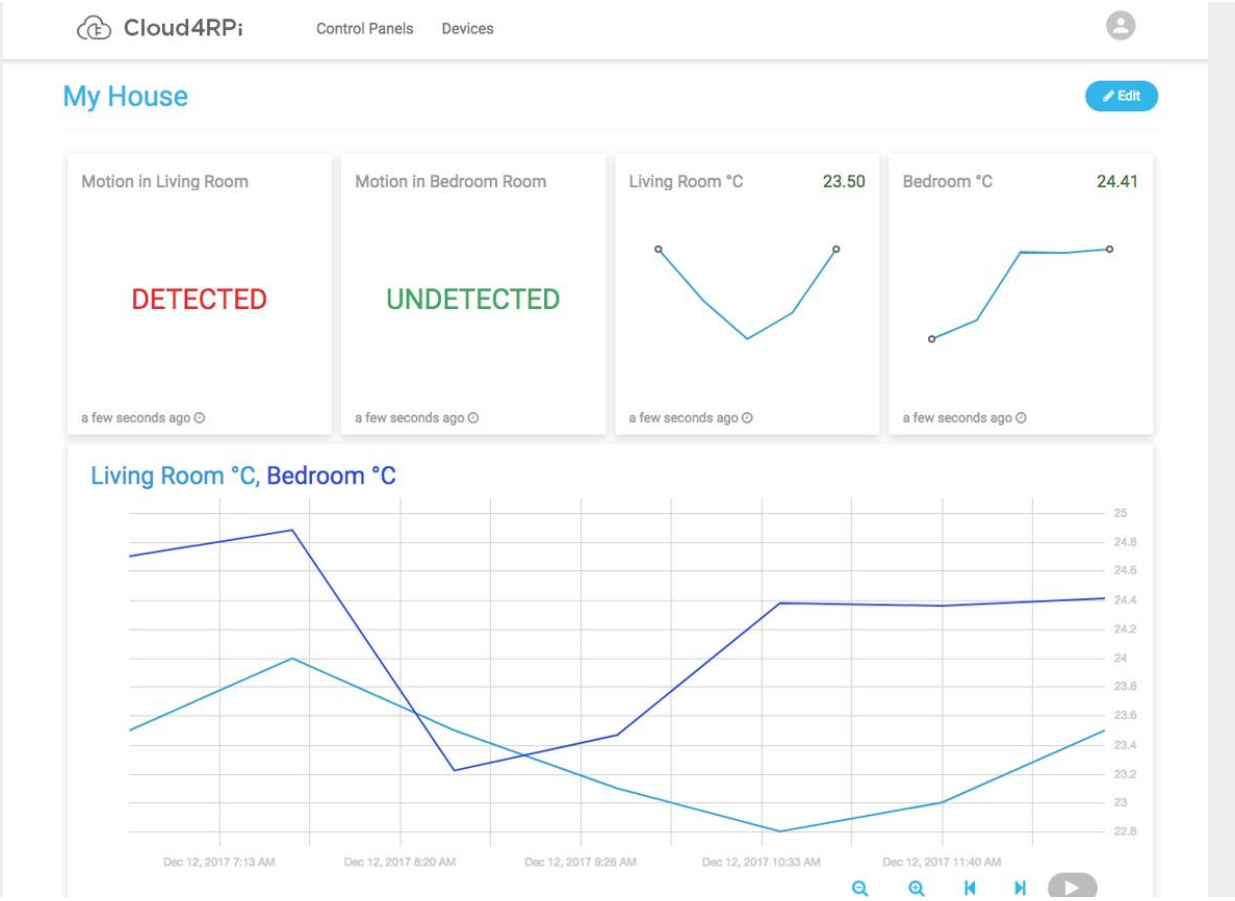
33. Smart Robot Car Dashboard
Display movement, distance, and sensor readings.



34. Smart Parking Occupancy Monitor
Display whether a parking space is occupied.



35. Interactive Robotic Display
Trigger animations or robot movements when someone approaches.



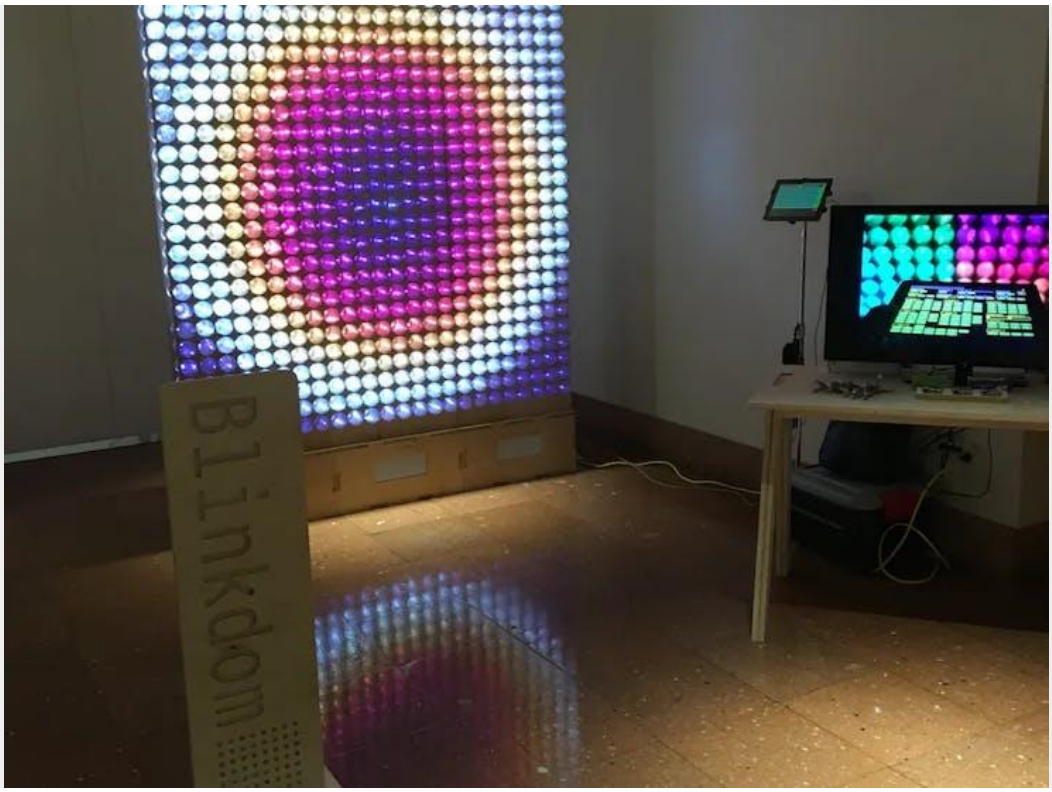
36. Multi-Sensor Monitoring Station
Display PIR, temperature, humidity, and light readings.



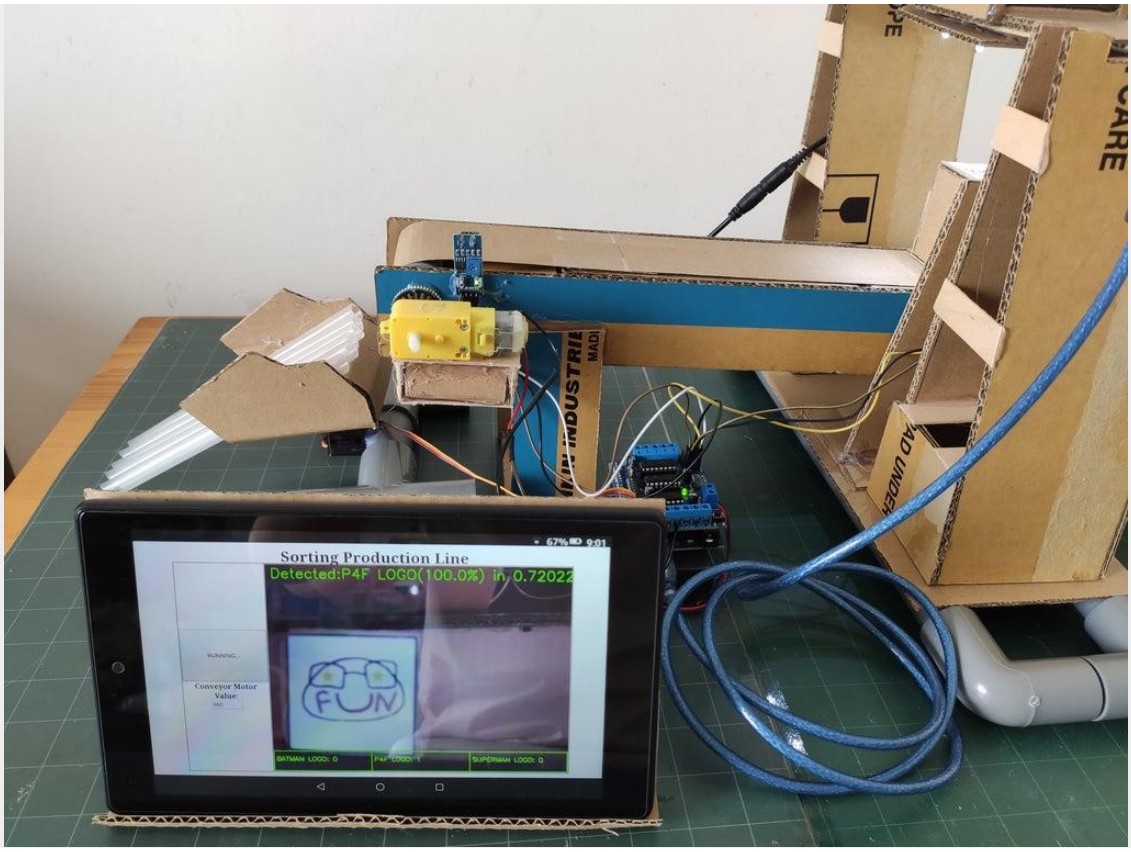
37. Smart Elevator Model
Detect approaching users and display elevator status.



38. Smart Vending Machine Interface
Wake the display when a customer approaches.



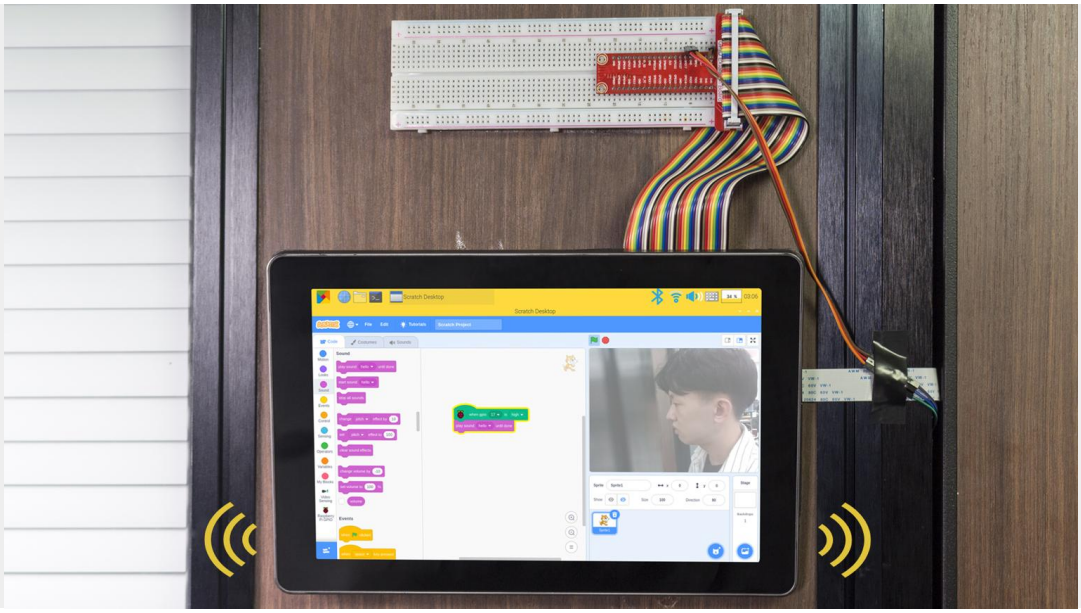
39. Interactive Art Installation
Change colors, animations, and graphics based on motion.



40. Smart Conveyor Monitor
Display detected activity and conveyor operating status.

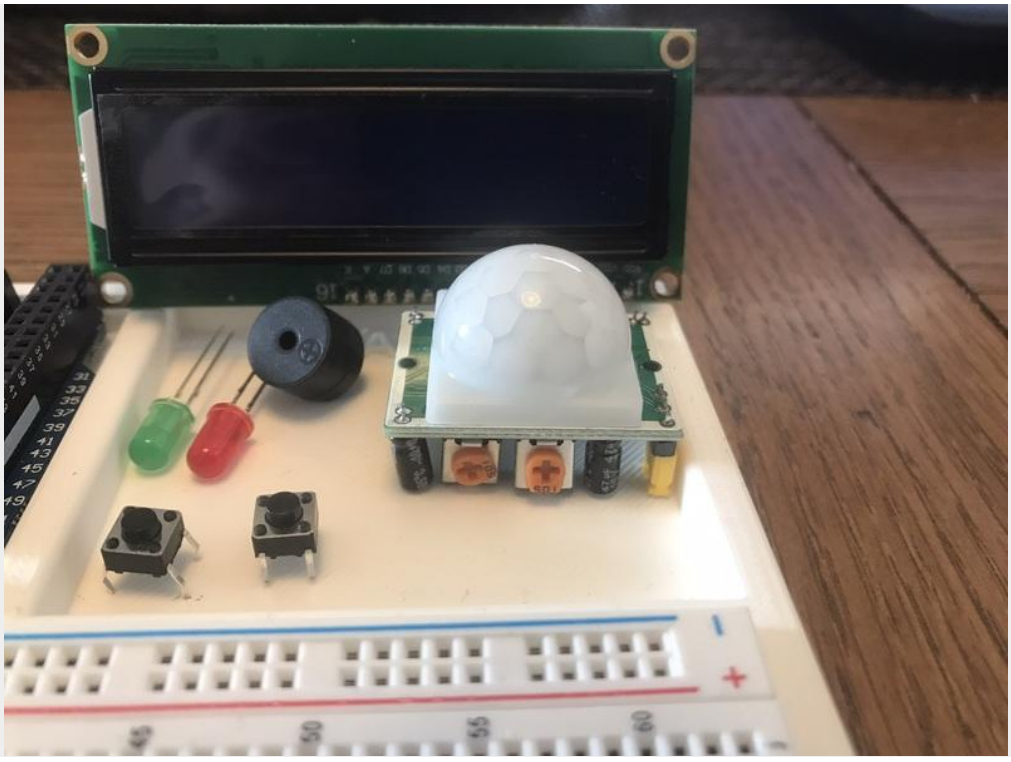
5. Advanced and creative projects

Projects 41–50



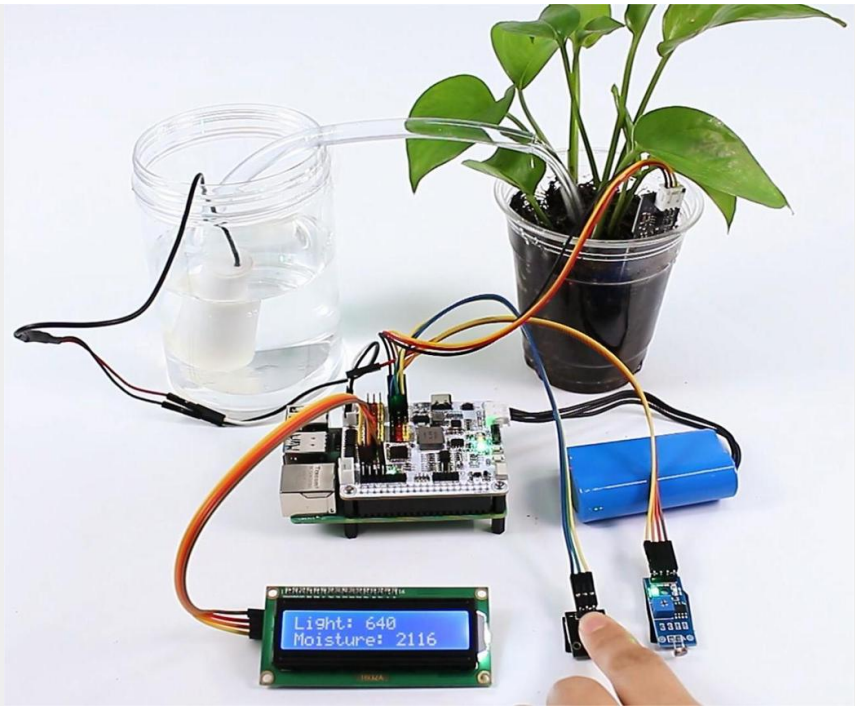
41. Motion-Activated Gaming Station

Wake a gaming interface when a player approaches.



42. Motion-Triggered Quiz Game

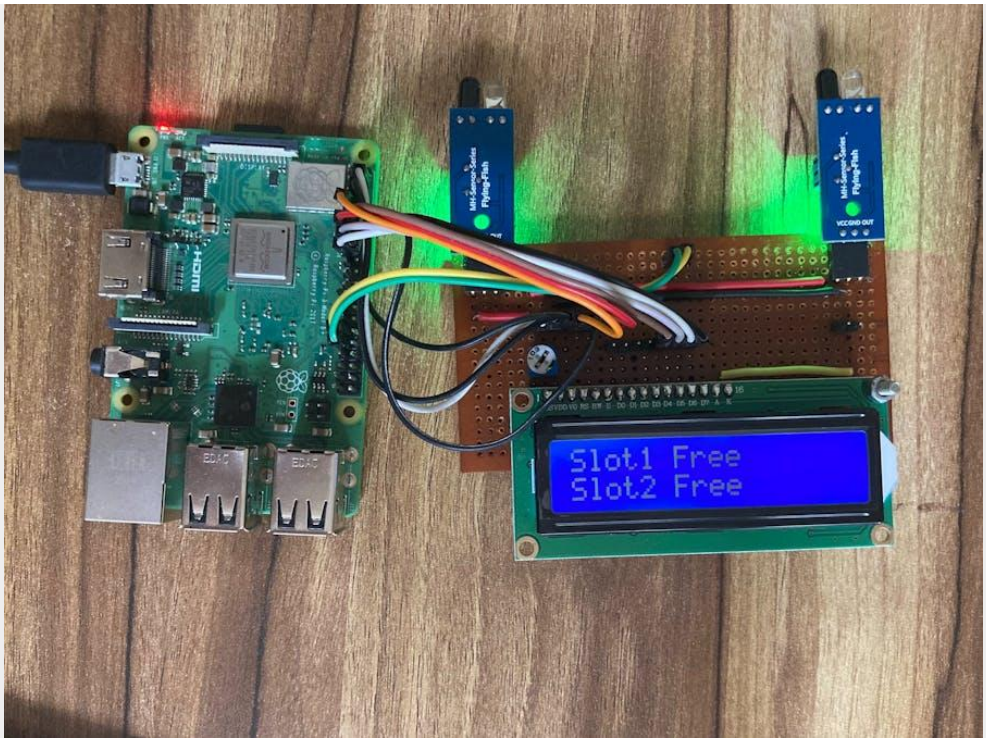
Display questions and start a quiz when someone approaches.



43. Smart Plant Monitoring System
Display soil moisture, temperature, and movement around plants.



44. Pet Activity Monitor
Detect pet activity and display captured images or activity counts.



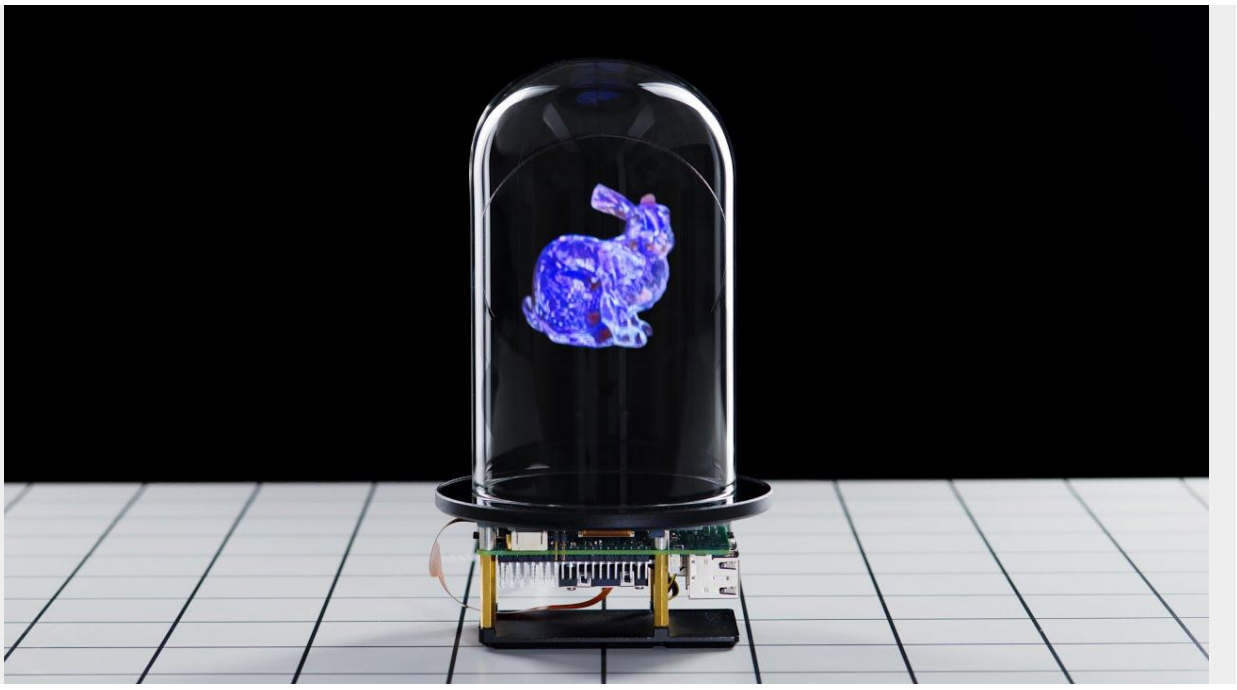
45. Smart Visitor Counter
Count motion-triggered entries and display visitor totals. Use two sensors for direction detection.



46. Smart Aquarium Monitor
Display aquarium conditions and detect activity nearby.



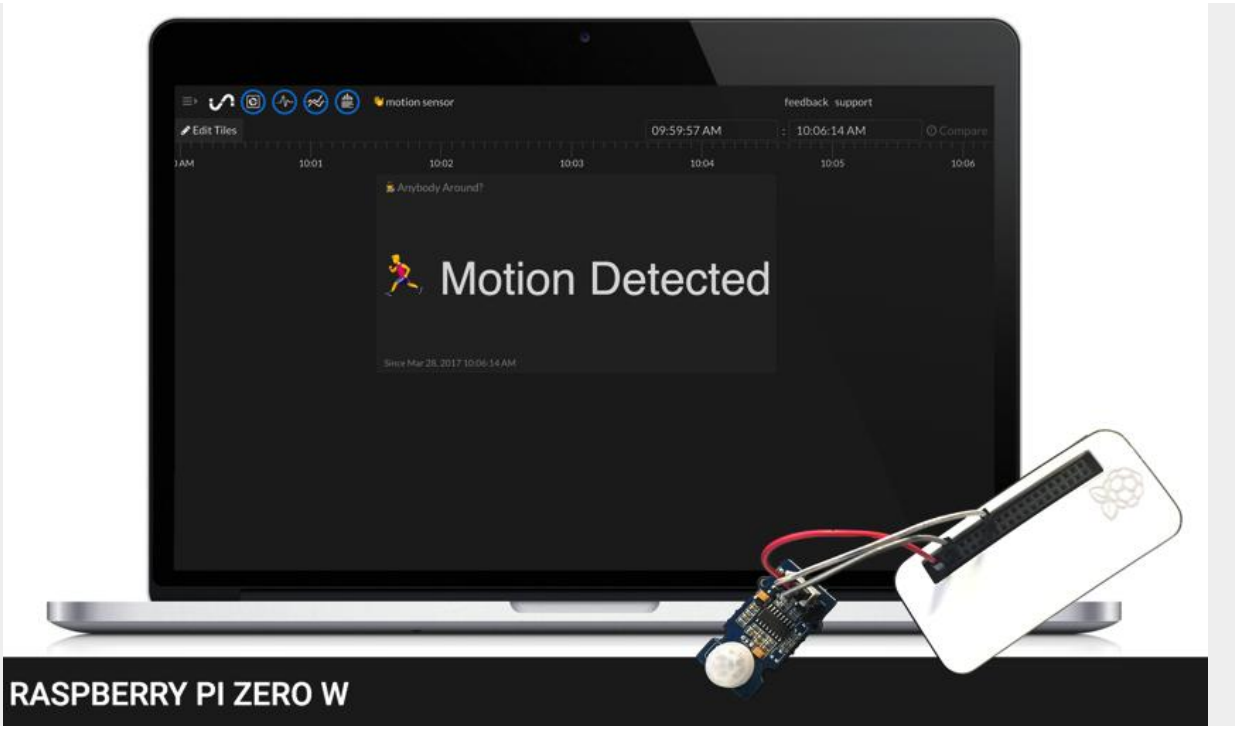
47. Smart Digital Advertising Board
Show advertisements when people approach.



48. Motion-Activated 3D Display
Activate a 3D-style animation or visual presentation when a visitor approaches.



49. Smart Classroom Monitor
Display occupancy, room status, and environmental readings.



50. AI-Powered Motion Recognition System
Combine a camera and AI to recognize people, pets, or objects and display detection results.

Hardware you can use



Raspberry Pi 4 or 5
Main computer for processing sensor data.



HC-SR501 PIR Motion Sensor
Detects movement from changes in infrared radiation.



3.5-inch or 5-inch TFT Display
For interactive graphical interfaces.



16×2 I2C LCD
For simple messages and sensor status.



0.96-inch OLED Display